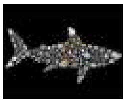
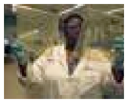
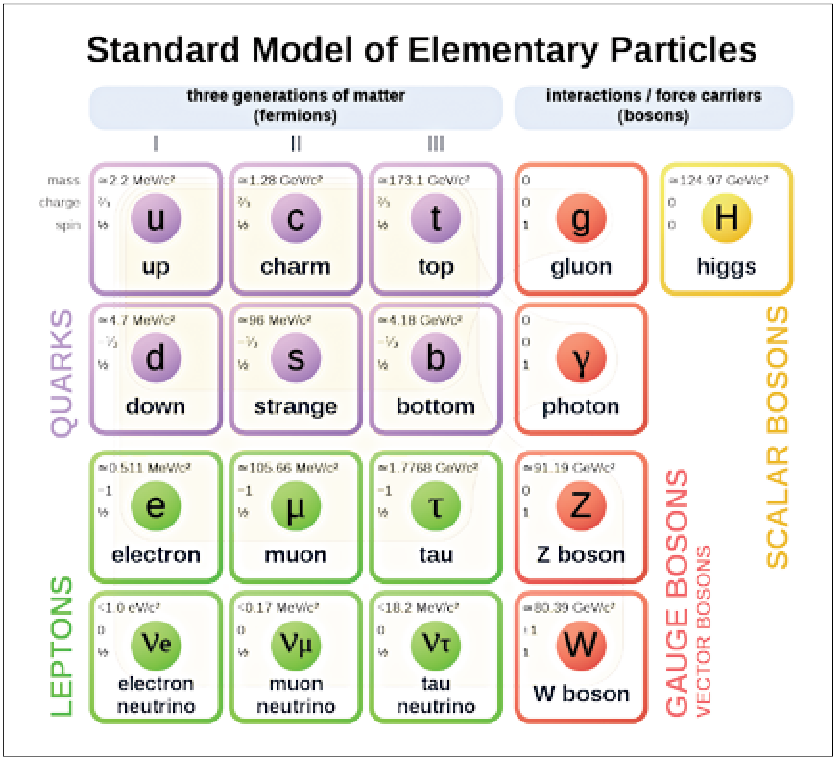


Shark-pocalypse?
Studies show that 19 million years ago, a vast majorities of sharks died off. https://dgit.in/aug21-39



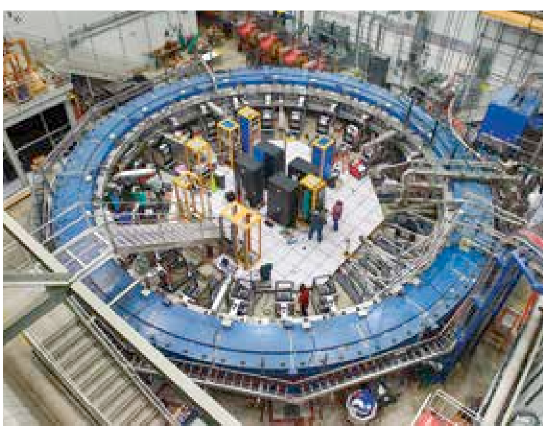
E. coli made virus-proof
Researchers successfully edited the genetic code of E. Coli bacteria to make it resistant to viruses. https://dgit.in/aug21-40

the higgsino, is lighter then it is a dark matter candidate, and if it is heavier then it is likely to decay into low momenta electrons and muons, making it difficult to be picked up by the detectors. Superpartners of the electrons and muons, sleptons may themselves decay into the LSP which would be at almost the same mass, making sleptons more difficult to detect. Then there is the stop, which is the supersymmetric partner of the top quark. Theoretically, the stop should have a mass range about the same as a top, and while tops have been spotted, no one has seen a stop yet. By not finding the stops where they expected to find them, researchers believe that they may find evidence of stops that have decayed into charm quarks, or even directly pick up signs of scharms. Researchers anticipate a squeezed mass spectrum for supersymmetric particles, and while the experiments at the LHC have not turned up any sleptons, stops, scharms or higgsinos, scientists have managed to put down increasing constraints on where they might be hiding, and what to look for in future experiments.



The Standard Model

In April 2021, the results of the Muon g-2 experiment were announced by BNL and Fermilab. These results were the culmination of a long series of experiments that began at CERN in the 1960s, and then by BNL between 1997 and 2001. The equipment was then shipped to Fermilab, where the equipment was improved to produce a more uniform magnetic field within the accelerator. The muons circulate around what is called the "storage ring" within the experiment, travelling nearly at the speed of light. During this time, scientists made very careful magnetic measurements of over 8 billion muons, finding out what is known as their g-factor. These were highly precise magnetic measurements of muons, cousins of electrons that are about 200 times as heavy, as they interacted with short lived virtual particles in the quantum foam. These precise measurements are very well predicted by the Standard Model, but if there are other forces and particles beyond the Standard Model in the universe, then these would cause a discrepancy in the measurements of the muons. In the Muon g-2 experiment, Fermilab confirmed the findings by BNL that showed that the muons did not behave in a way that was completely explained by the Standard Model. The combined results of both the laboratories show a difference with the expected theory at a significance of 4.2 sigma, which is just shy of the 5 sigma needed to announce the discovery, but the finding still is considered compelling evidence of physics beyond the Standard Model, with the chances that the results are a statistical fluctuation being one in forty thousand. Only the results from the first run are out, and there are four more runs left.



The Muon g-2 storage ring at Fermilab

Theoretical physics beyond the standard model is a dark frontier, but one that emerges from equations and improved computational and statistical methods. Supersymmetry is just one category of theoretical predictions for physics beyond the standard model that can explain some of its shortcomings. There are whole clusters of other theories including the Little Higgs theories (the one we have spotted is one of many Higgs particles) and explanations that involve additional dimensions. d